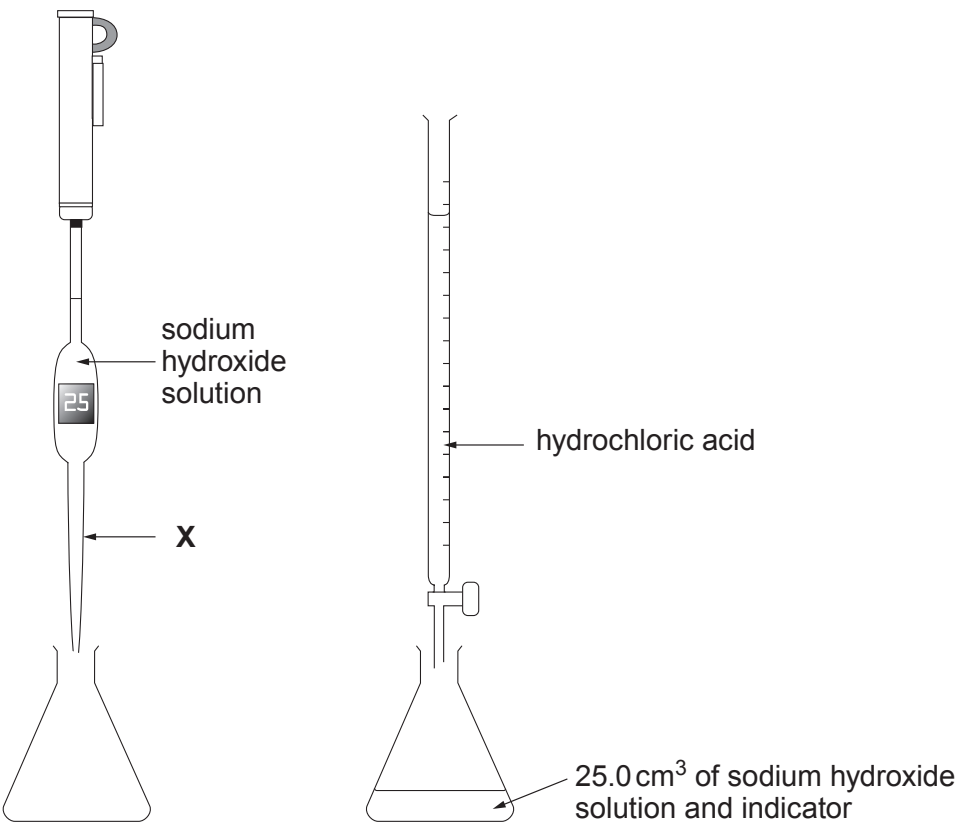


7. (a) Hydrochloric acid, HCl, reacts with sodium hydroxide solution to form sodium chloride and water only.

Write a balanced **symbol** equation for this reaction. [2]

- (b) A group of students was asked to find the volume of hydrochloric acid solution needed to neutralise 25.0 cm<sup>3</sup> of sodium hydroxide solution. They decided to titrate sodium hydroxide with hydrochloric acid.

Apparatus



Results

| Titre                                                 | 1    | 2    | 3    |
|-------------------------------------------------------|------|------|------|
| Volume of hydrochloric acid needed (cm <sup>3</sup> ) | 18.2 | 17.8 | 18.0 |



- (i) Name the piece of apparatus **X**. [1]

.....

- (ii) Explain the purpose of the indicator. [1]

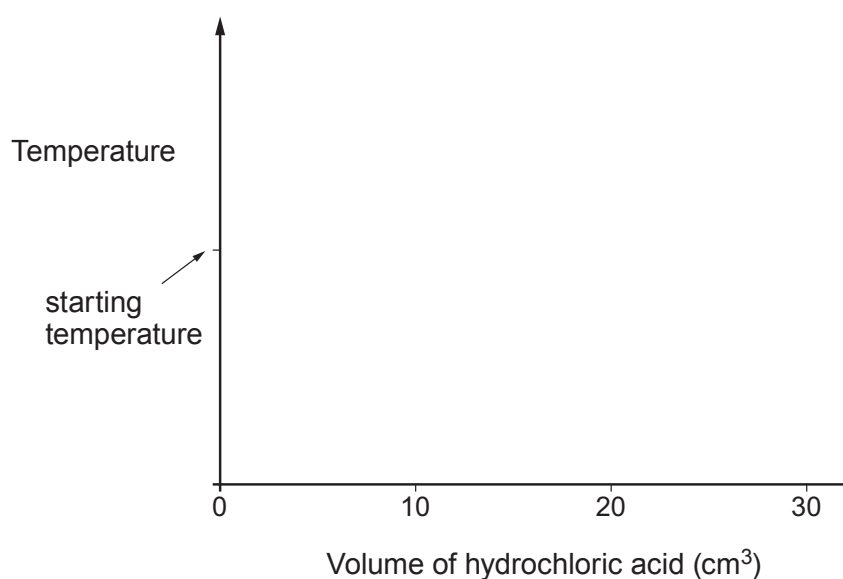
.....

- (iii) Calculate the mean volume of hydrochloric acid needed to neutralise  $25.0\text{ cm}^3$  of the sodium hydroxide solution. [1]

Mean volume = .....  $\text{cm}^3$

- (iv) The change in temperature during the reaction can be monitored using a temperature sensor.

Sketch a graph on the axes below to show how the temperature changes as more and more acid up to a total of  $30\text{ cm}^3$  is added. [2]



- (v) The experiment was repeated using hydrochloric acid of **half** the original concentration.

State the volume of hydrochloric acid that would be needed to change the indicator colour. [1]

Volume = .....  $\text{cm}^3$

